

Bluestock Mutual Fund Analytics

Final Project Report

Fund Performance, Risk, Investor Behaviour & Interactive Dashboard

Prepared for Bluestock Fintech

August 2026

1. Executive Summary

This capstone project develops an end-to-end Mutual Fund Analytics solution covering data ingestion, cleaning, exploratory analysis, performance measurement, risk analytics, investor behaviour, portfolio concentration and interactive Power BI reporting. The solution combines Python, Pandas, NumPy, SQL and Power BI to transform multiple mutual-fund datasets into decision-ready outputs.

The analysis covers 40 schemes and integrates fund master data, NAV history, AUM, SIP inflows, category flows, investor transactions, benchmark indices and portfolio holdings. The final analytical layer produces historical VaR/CVaR, rolling 90-day Sharpe analysis, investor cohorts, SIP continuity monitoring, a risk-based fund recommender and sector HHI concentration analysis.

Key findings: SBI Small Cap Fund - Direct Plan - Growth recorded the most negative 95% VaR among the supplied results at -2.69%, with CVaR of -3.24%. The 2024 investor cohort contributed approximately ₹225.81 crore in total investment, while the 2025 cohort had the higher average SIP amount of approximately ₹13,505. SIP continuity analysis classified 97.8% of eligible investors as at-risk under the defined 35-day average-gap rule.

The dashboard contains four pages: Industry Overview, Fund Performance & Risk Analysis, Investor & Transaction Analytics, and SIP & Market Trends.

2. Problem Statement & Objectives

Mutual-fund information is distributed across multiple datasets and granularities. The objective is to build a repeatable analytical workflow that converts cleaned data into reliable performance, risk and investor insights and presents them through an executive-friendly dashboard.

- Integrate and clean multiple mutual-fund datasets.
- Calculate daily returns and fund performance metrics.
- Compare funds against benchmark indices.
- Analyze investor transactions, cohorts and SIP continuity.
- Measure downside risk using VaR and CVaR.
- Measure risk-adjusted performance using rolling Sharpe ratios.
- Measure portfolio sector concentration using HHI.
- Build an interactive Power BI dashboard and professional project documentation.

3. Data Sources

The project uses cleaned versions of the mutual-fund datasets in the processed-data layer. These include fund master, NAV history, AUM, benchmark indices, category inflows, monthly SIP inflows, industry folio counts, investor transactions, portfolio holdings and scheme performance.

Dataset	Purpose
Fund Master	Scheme identity, category, plan and risk metadata
NAV History	Historical NAV and daily return calculations
AUM	Industry and fund-house asset trends
Benchmark Indices	Benchmark return comparison
Category Inflows	Category-level net-flow analysis
Monthly SIP Inflows	SIP trend analysis

Industry Folio Count	Folio/industry indicators
Investor Transactions	Investor-level transaction and cohort analysis
Portfolio Holdings	Sector weights and concentration
Scheme Performance	Return and risk-adjusted metrics

4. ETL & Data Processing Design

The project follows a staged ETL approach: raw data is ingested, validated, cleaned and written to processed datasets before analytical notebooks and Power BI consume the results. Dates are standardized, duplicate records and missing values are reviewed, and numeric fields are converted to appropriate types.

- Raw data → validation → cleaning → processed datasets
- Python/Pandas for transformation and advanced analytics
- SQL for structured analytical queries
- Power BI for interactive reporting
- Separate analytical outputs for reproducibility

5. Exploratory Data Analysis

5.1 Data Quality

- Historical NAV observations were standardized by scheme and date.
- Daily returns were checked for reasonable distributions and extreme observations.
- Fund categories and risk classifications were compared.
- Investor transactions were segmented by transaction type and cohort.
- Portfolio holdings were examined by sector and weight.

5.2 Dashboard EDA Highlights

The dashboard provides an executive view of industry AUM, fund performance, investor transactions and SIP/market trends. The following screenshots are the actual dashboard pages supplied for this final report.

6. Fund Performance Analysis

Daily returns form the base for performance and risk calculations. The rolling 90-day Sharpe ratio uses rolling mean daily return divided by rolling daily standard deviation and annualizes the result using $\sqrt{252}$. The Fund Performance page combines return-versus-risk analysis, NAV trends and benchmark-indexed performance.

Industry Overview



63

Total AUM (₹ L Cr)

940K

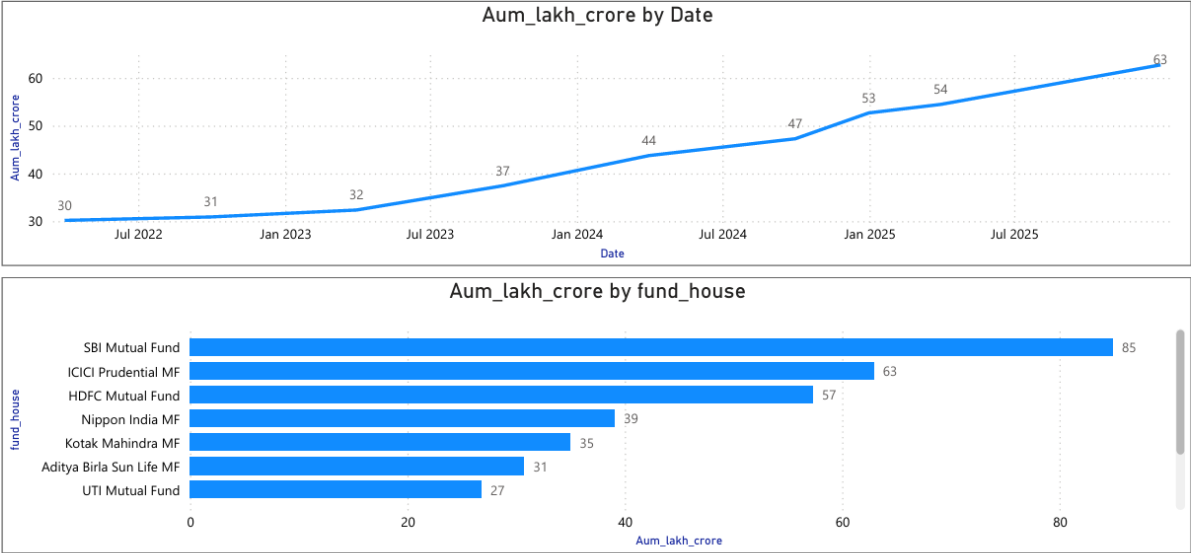
Total SIP Inflows

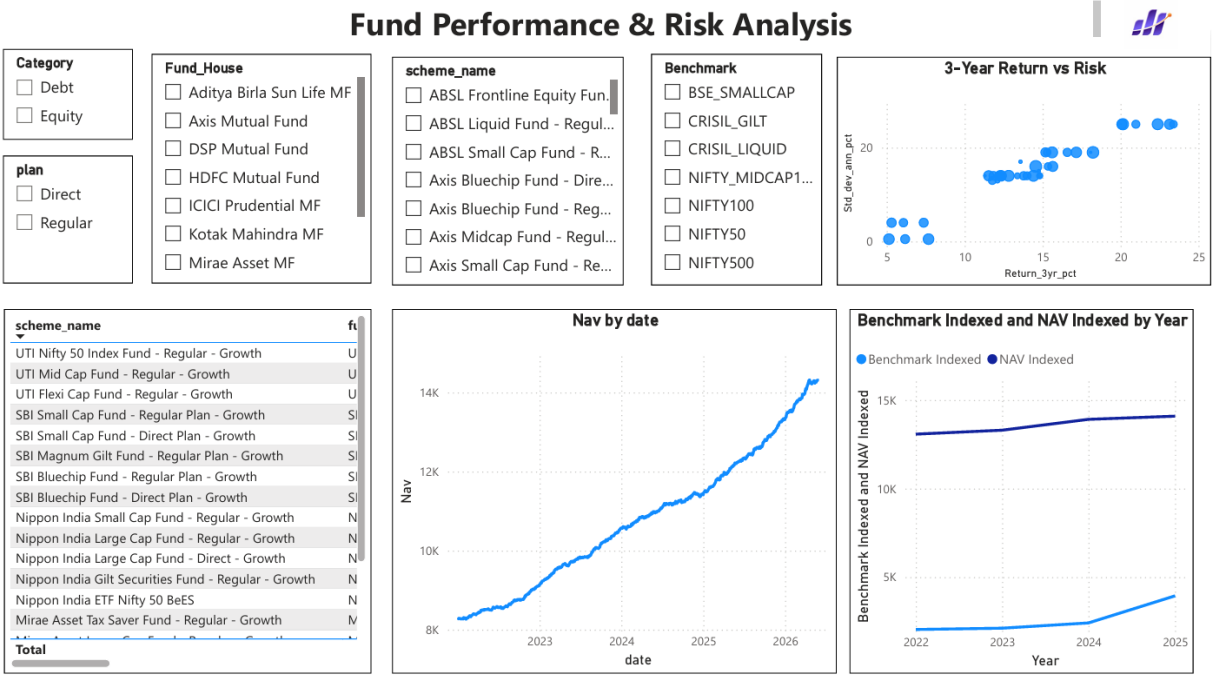
26.12

Total Folios

14K

Total Schemes





Investor & Transaction Analytics



city_tier

☐ B30

☐ T30

Transaction_Type Distribution

0bn (6.17%)

1bn (35....)

2bn (58.49%)

transaction_type

Lumpsum

Redemption

SIP

Average SIP Amount by Age Group

Average of amount_inr

0.0M

0.1M

18-25

26-35

46-55

36-45

56+

age_group

Transaction Amount by State

state

Punjab

Tamil Nadu

Madhya Pradesh

Rajasthan

Gujarat

West Bengal

Telangana

Delhi

Uttar Pradesh

Haryana

Karnataka

Maharashtra

Amount_inr

0.0bn

0.1bn

0.2bn

0.3bn

Monthly Transaction Volume

Transaction_date

40

60

80

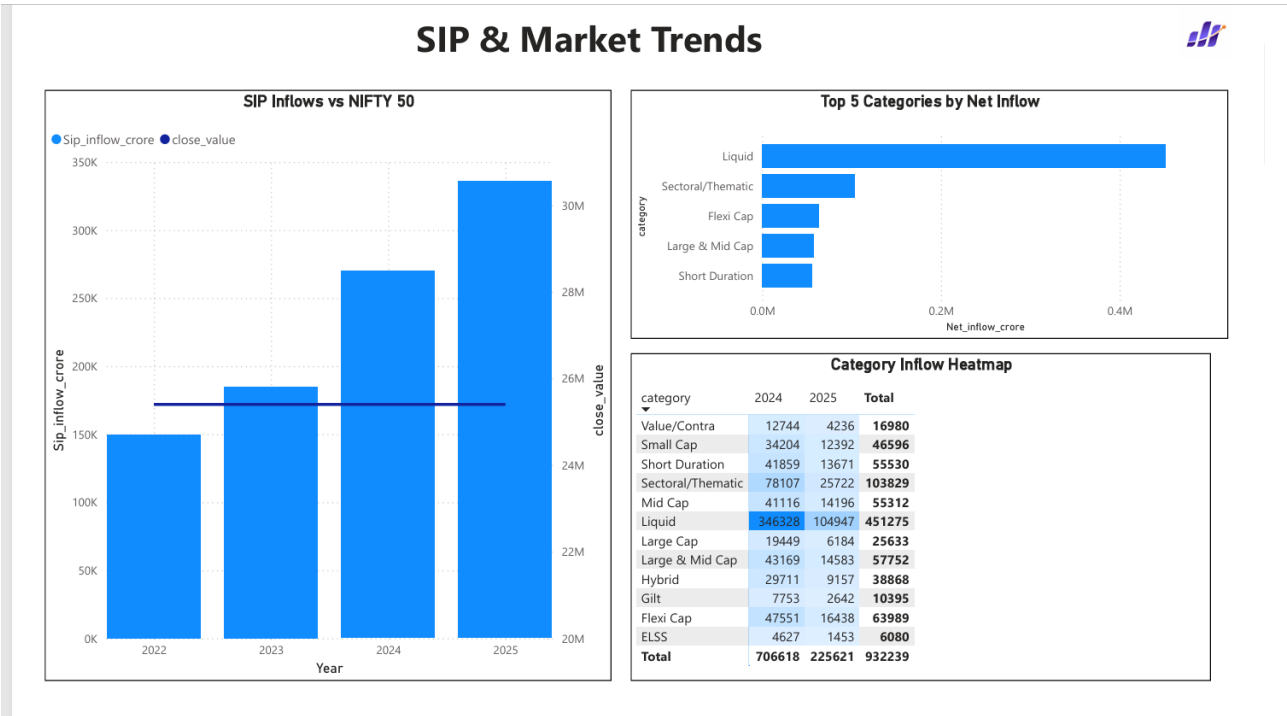
Jan 2024

Jul 2024

Jan 2025

Year

Investor & Transaction Analytics — transaction mix, SIP behaviour, state-wise amount and monthly transaction volume.



SIP & Market Trends — SIP inflows versus NIFTY 50, category net inflow ranking and category heatmap.

9. Key Findings

- Small-cap Very High-risk funds dominate the most negative VaR observations.
- The 2024 cohort contributed substantially more total investment than the 2025 cohort despite the 2025 cohort having a higher average SIP amount.
- 97.8% of eligible investors were classified as at-risk under the 35-day SIP-gap rule.
- Risk-based fund recommendations differ materially across Low, Moderate and High risk grades.
- Axis Bluechip Fund - Regular - Growth showed the highest HHI among the displayed equity-fund results.

10. Recommendations

- Use VaR/CVaR alongside return metrics when evaluating downside exposure.
- Monitor SIP-gap behaviour as an early-warning indicator for investor engagement.
- Use risk-adjusted metrics such as Sharpe rather than return alone.
- Review highly concentrated portfolios for sector concentration exposure.
- Provide risk-appetite-based fund shortlists instead of one-size-fits-all recommendations.
- Refresh the dashboard periodically so current NAV, flows and investor activity remain decision-relevant.

11. Limitations

- Historical VaR/CVaR are backward-looking.
- Sharpe depends on the selected time window and standard deviation as a risk proxy.
- The 35-day SIP threshold is a project-defined rule.

- HHI measures concentration but not inter-sector correlation.
- The recommender is an analytical ranking and not financial advice.
- Dashboard results reflect the project dataset and available date range.

12. Technical Deliverables & Final Review

- Advanced_Analytics.ipynb
- var_cvar_report.csv
- recommender.py
- rolling_sharpe_chart.png
- bluestock_mf_dashboard.pbix
- Dashboard.pdf and four dashboard PNGs
- Final_Report.pdf
- Bluestock_MF_Presentation.pptx
- README.md and v1.0 Git tag

Final review checklist: objectives covered, analytical outputs generated, dashboard pages documented, report prepared, presentation prepared, code cleaned, README prepared, and repository ready for final version tagging.

13. Conclusion

The Bluestock Mutual Fund Analytics capstone demonstrates an end-to-end workflow from multi-source data preparation to advanced risk metrics, investor behaviour analysis and interactive Power BI reporting. The final dashboard and analytical outputs provide a practical foundation for fund comparison, downside-risk monitoring, investor retention analysis and portfolio concentration review.